



SCM755 Lean Six Sigma

Challenge: Production growing 2-3% weekly, outpacing packaging capacity

Goal: Increase packing rate from 50-60 to 70 pouches/hour/person

DMAIC Process



Define
Identify problem worth solving now



Analyze
Find root causes through data



Control
Sustain gains through standardized processes



Measure
Collect data showing current performance



Improve
Implement and test solutions systematically

Define Phase

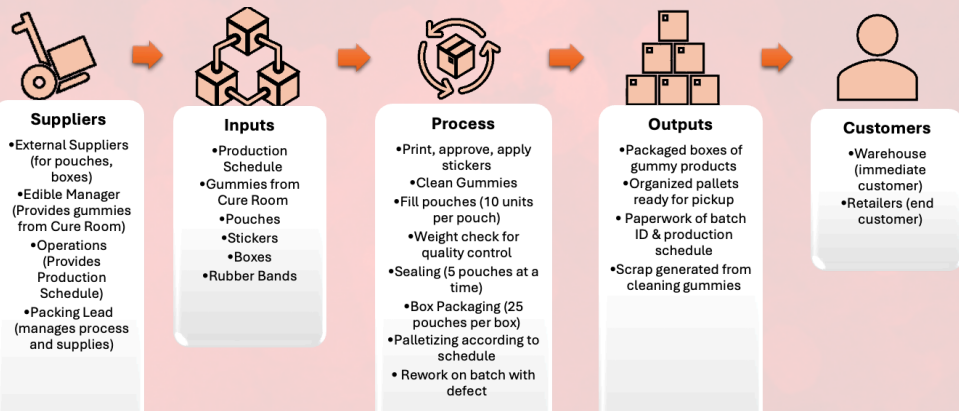
The Problem

Production volume increasing 2-3% weekly
Current packaging rate: 50-60 pouches/hour/person
Need to increase efficiency without more labor

Project Charter

Scope: Manual packing process optimization only
Out of Scope: Supply chain redesign, product formulation
Team: Lean Six Sigma practitioners + frontline workers

SIPOC Diagram

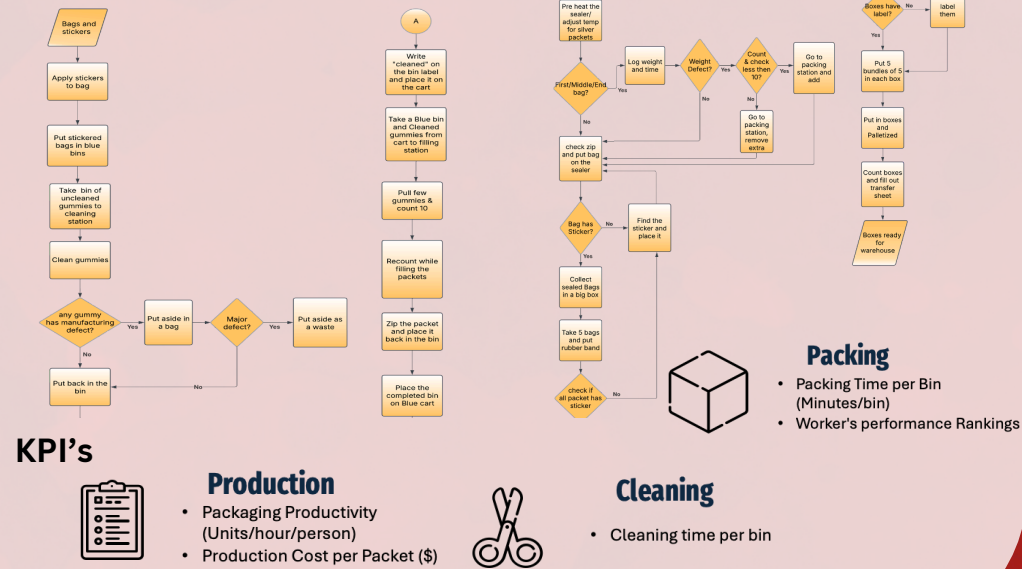


Measure Phase

Data Collection Method

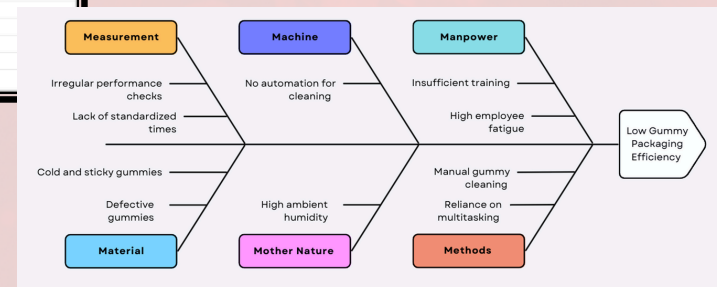
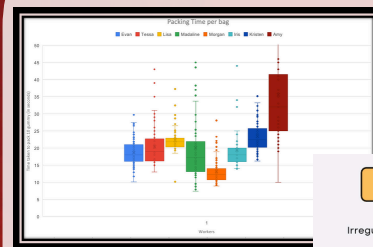
- 9 workers observed
- 120 samples collected
- Time studies for cleaning, packing, stickering

Detailed Flow Chart



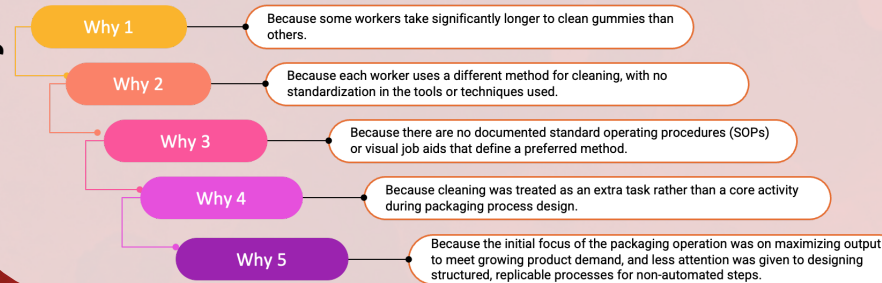
Analyze Phase

Time Study



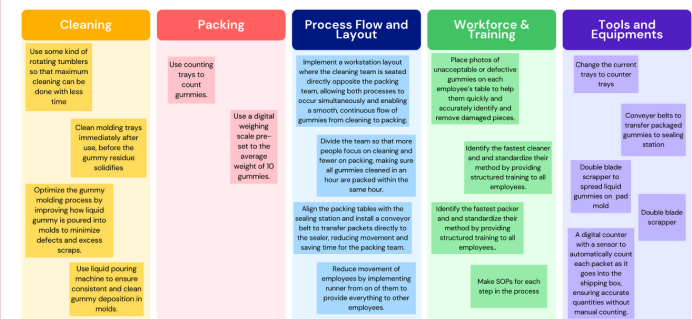
Problem Statement: Cleaning times vary greatly across workers, leading to inconsistent throughput and delays in downstream processes.

5 Why's

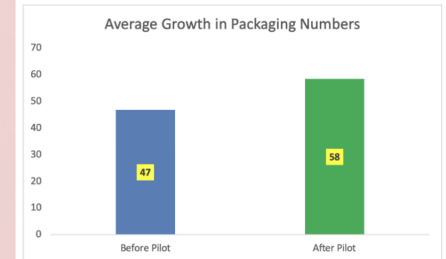
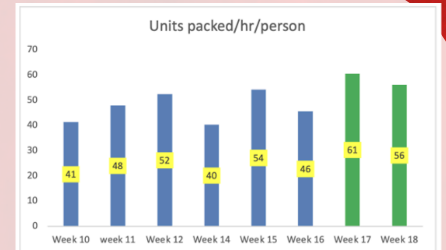
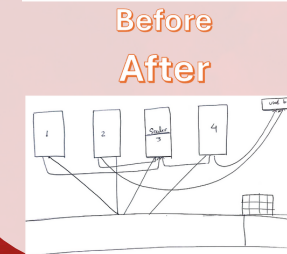


Improve Phase

Affinity Diagram



Spaghetti Diagram

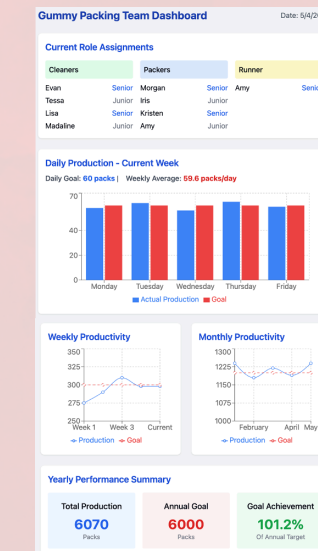


Savings



Control Phase

Dashboard



Defined Roles

	Primary Responsibility	Key Duties	Expected Outcome
Cleaner	Clean and inspect gummies before forwarding to packing	Receive raw gummies, clean thoroughly, inspect for defects, separate defective pieces	Consistent delivery of quality-approved gummies with minimal movement
Packer	Pack cleaned gummies with speed and accuracy	Count exactly 10 gummies per bag, seal bags properly, alert runner when supplies low	Consistent packing with accurate counts and properly sealed package
Runner	Support workflow by delivering materials to eliminate station leaving	Transport materials between stations, perform stickering, ensure correct packet types	Uninterrupted production flow with minimal delays and accurate material delivery
Team Lead	Oversee shift operations and assign roles based on employee strengths	Assign workers strategically, monitor productivity, resolve workflow issues, track KPIs	Well-balanced team with sustained productivity and quick issue resolution

Before-After

Metric	Before	After	Improvement
Task Allocation	Everyone doing the same task	Split roles based on individual speed	Better balance and output
Worker Movement	High (each worker fetched boxes)	Zero (runner delivers to stations)	Reduced idle time & fatigue
Output per worker/hour	~47 boxes	~58 boxes	23.5%

Performance Monitoring System

PACKING						
	Time In	Time out	Difference	Rank	Time taken	Bins per hour
Person 1	1:10 PM	1:50 PM	0:40	5	0:67	1.50
Person 2	5:35 AM	6:45 AM	1:10	7	1:17	0.86
Person 3	3:00 PM	5:45 PM	2:45	8	2:75	0.36
Person 4	6:52 PM	7:55 PM	1:03	6	1:05	0.95
Person 5	7:50 PM	8:55 PM	1:05	6	1:08	0.92
Person 6	2:52 AM	3:45 AM	0:53	3	0:86	1.13
Person 7	6:30 AM	7:20 AM	0:50	2	0:83	1.20
Person 8	10:19 AM	11:15 AM	0:56	4	0:93	1.07